Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Garcia's Registered Jerseys

Physical address of dairy:

Number and Street	City	County	Zip Code
1018 Hickman RD	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 08/15/2002

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X024-X008-X017-XXXX

B. OPERATORS

Garcia, Barbara Olene

Operator name: Garcia, Barbara Olene

Telephone no.: (209) 669-7838 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
PO Box 544	Denair	CA	95316

This operator is responsible for paying permit fees.

Garcia, Jose Luis

Operator name: Garcia, Jose Luis

Telephone no.: (209) 669-7838 (Landline) (209) 678-2044 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 544	Denair	CA	95316

C. OWNERS

Garcia, Barbara Olene

Legal owner name:	Garcia, Barbara Olene	Telephone no.:	(209) 669-7838	
			Landline	Cellular
PO Box 544	Denair		CA	95316
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

Garcia, Jose Luis

Legal owner name:	Garcia, Jose Luis	Telephone no.:	(209) 669-7838	(209) 678-2044
			Landline	Cellular
P.O. Box 544	Denair		CA	95316
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	550	570	230	0
Number under roof	1,390	90	50	0	0	0
Maximum number	1,400	95	600	600	250	0
Average number	1,400	95	560	600	225	0
Avg live weight (lbs)	1,100	1,100	750	550		

Predominant milk cow breed: Jersey

Average milk production: 54 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 46,081.13 *tons per reporting period*

Total nitrogen from manure: 574,079.24 *lbs per reporting period*

After ammonia losses (30% loss applied): 401,855.47 *lbs per reporting period*

Total phosphorus from manure: 94,900.11 *lbs per reporting period*

Total potassium from manure: 253,396.12 *lbs per reporting period*

Total salt from manure: 681,035.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,985,500 *gallons*

Total nitrogen generated: 50,231.48 *lbs*

Total phosphorus generated: 6,187.70 *lbs*

Total potassium generated: 95,582.80 *lbs*

Total salt generated: 411,816.57 *lbs*

	7,985,500	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	7,985,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	13	13	2	process wastewater	X024-X008-X017-XXXX
Field 2	38	38	1	process wastewater	X024-X008-X017-XXXX
Field 3	56	56	2	process wastewater	X024-X004-X024-XXXX X024-X008-X017-XXXX
Field 4	72	72	2	process wastewater	X024-X004-X024-XXXX
Totals for areas that were used for application	179	179	7		
Totals for areas that were not used for application					
Land application area totals	179	179	7		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 13 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	92.00 *ton*	As-is		58.8	8,090.00	1,130.00	11,800.00		16.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	7.08	114.50	15.99	167.02	962.18

Field 1

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/27/2023	206.00 *ton*	As-is		68.4	4,490.00	570.00	8,310.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	15.85	142.30	18.06	263.36	781.15

Field 2

Field name: Field 2

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/27/2023	517.00 *ton*	As-is		56.8	5,100.00	570.00	5,770.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	13.61	138.77	15.51	157.00	728.81

Field 3

Field name: Field 3

Field 3

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	264.00 *ton*	As-is		60.4	7,650.00	1,100.00	10,700.00		13.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	6.95	106.29	15.28	148.67	715.30

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 56 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	741.00 *ton*	As-is		66.3	5,160.00	680.00	8,730.00		7.75

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	13.23	136.56	18.00	231.03	691.18

Field 4

Field name: Field 4

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	230.98 *ton*	As-is		62.4	7,240.00	990.00	9,100.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	7.45	107.89	14.75	135.61	661.17

Field 4

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 72 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,046.00 *ton*	As-is		65.9	4,810.00	400.00	8,640.00		7.58

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	14.53	139.76	11.62	251.04	751.02

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/27/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	118.92	19.98	396.39	935.47	123,500.00 *gal*
Application event totals		118.92	19.98	396.39	935.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	2.33	0.00	0.00	173.32	2,160,000.00 *gal*
Application event totals		2.33	0.00	0.00	173.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.13	7.33	128.06	826.16	150,000.00 *gal*
Ag Well	Ground water	0.19	0.00	0.00	14.44	180,000.00 *gal*
Application event totals		64.32	7.33	128.06	840.60	

Field 1 - 06/29/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 1 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	102.26	13.63	146.74	785.71	180,000.00 *gal*
Ag Well	Ground water	2.02	0.00	0.00	150.21	1,872,000.00 *gal*
Application event totals		104.28	13.63	146.74	935.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.28	4.74	79.25	419.82	150,000.00 *gal*
Ag Well	Ground water	1.71	0.00	0.00	127.10	1,584,000.00 *gal*
Application event totals		53.99	4.74	79.25	546.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.63	0.00	0.00	121.32	1,512,000.00 *gal*
Application event totals		1.63	0.00	0.00	121.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.55	0.00	0.00	115.55	1,440,000.00 *gal*
Application event totals		1.55	0.00	0.00	115.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.28	4.74	79.25	419.82	150,000.00 *gal*
Ag Well	Ground water	1.40	0.00	0.00	103.99	1,296,000.00 *gal*
Application event totals		53.68	4.74	79.25	523.81	

Field 1 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.40	0.00	0.00	103.99	1,296,000.00 *gal*
Application event totals		1.40	0.00	0.00	103.99	

10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.40	0.00	0.00	103.99	1,296,000.00 *gal*
Application event totals		1.40	0.00	0.00	103.99	

Field 2 - 06/29/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.29	12.44	133.87	716.79	480,000.00 *gal*
Ag Well	Ground water	2.28	0.00	0.00	169.97	6,192,000.00 *gal*
Application event totals		95.57	12.44	133.87	886.77	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.24	5.19	86.75	459.59	480,000.00 *gal*
Ag Well	Ground water	2.18	0.00	0.00	162.07	5,904,000.00 *gal*
Application event totals		59.42	5.19	86.75	621.66	

Field 2 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	2.07	0.00	0.00	154.16	5,616,000.00 *gal*
Application event totals		2.07	0.00	0.00	154.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	2.02	0.00	0.00	150.21	5,472,000.00 *gal*
Application event totals		2.02	0.00	0.00	150.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.08	4.54	75.91	402.14	420,000.00 *gal*
Ag Well	Ground water	1.99	0.00	0.00	148.23	5,400,000.00 *gal*
Application event totals		52.08	4.54	75.91	550.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.97	0.00	0.00	146.26	5,328,000.00 *gal*
Application event totals		1.97	0.00	0.00	146.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.91	0.00	0.00	142.30	5,184,000.00 *gal*
Application event totals		1.91	0.00	0.00	142.30	

Field 3 - 10/27/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 3 - 10/27/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	118.92	19.98	396.39	935.47	361,000.00 *gal*
Application event totals		118.92	19.98	396.39	935.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	2.07	0.00	0.00	154.16	5,616,000.00 *gal*
Application event totals		2.07	0.00	0.00	154.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.88	5.01	87.62	565.26	300,000.00 *gal*
Ag Well	Ground water	0.16	0.00	0.00	11.86	432,000.00 *gal*
Application event totals		44.04	5.01	87.62	577.12	

Field 3 - 06/20/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.95	12.66	136.26	729.59	720,000.00 *gal*
Ag Well	Ground water	2.13	0.00	0.00	158.26	8,496,000.00 *gal*
Application event totals		97.08	12.66	136.26	887.85	

Field 3 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.26	5.28	88.30	467.80	720,000.00 *gal*
Ag Well	Ground water	2.02	0.00	0.00	150.21	8,064,000.00 *gal*
Application event totals		60.28	5.28	88.30	618.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.98	0.00	0.00	147.53	7,920,000.00 *gal*
Application event totals		1.98	0.00	0.00	147.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.95	0.00	0.00	144.85	7,776,000.00 *gal*
Application event totals		1.95	0.00	0.00	144.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.55	4.40	73.59	389.83	600,000.00 *gal*
Ag Well	Ground water	1.91	0.00	0.00	142.16	7,632,000.00 *gal*
Application event totals		50.46	4.40	73.59	531.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.89	0.00	0.00	140.82	7,560,000.00 *gal*
Application event totals		1.89	0.00	0.00	140.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.82	0.00	0.00	135.46	7,272,000.00 *gal*
Application event totals		1.82	0.00	0.00	135.46	

Field 4 - 10/27/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	145.77	24.49	485.89	1,146.71	361,000.00 *gal*
Application event totals		145.77	24.49	485.89	1,146.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	2.21	0.00	0.00	164.75	4,896,000.00 *gal*
Application event totals		2.21	0.00	0.00	164.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.89	3.07	53.70	346.45	150,000.00 *gal*
Ag Well	Ground water	0.08	0.00	0.00	6.06	180,000.00 *gal*
Application event totals		26.97	3.07	53.70	352.51	

Field 4 - 06/20/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.47	13.13	141.31	756.61	960,000.00 *gal*
Ag Well	Ground water	2.13	0.00	0.00	158.56	10,944,000.00 *gal*
Application event totals		100.60	13.13	141.31	915.17	

Field 4 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.87	4.79	80.13	424.48	840,000.00 *gal*
Ag Well	Ground water	2.02	0.00	0.00	150.21	10,368,000.00 *gal*
Application event totals		54.88	4.79	80.13	574.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.96	0.00	0.00	146.04	10,080,000.00 *gal*
Application event totals		1.96	0.00	0.00	146.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.91	0.00	0.00	141.87	9,792,000.00 *gal*
Application event totals		1.91	0.00	0.00	141.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.87	4.79	80.13	424.48	840,000.00 *gal*
Ag Well	Ground water	1.93	0.00	0.00	143.95	9,936,000.00 *gal*
Application event totals		54.80	4.79	80.13	568.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.85	0.00	0.00	137.69	9,504,000.00 *gal*
Application event totals		1.85	0.00	0.00	137.69	

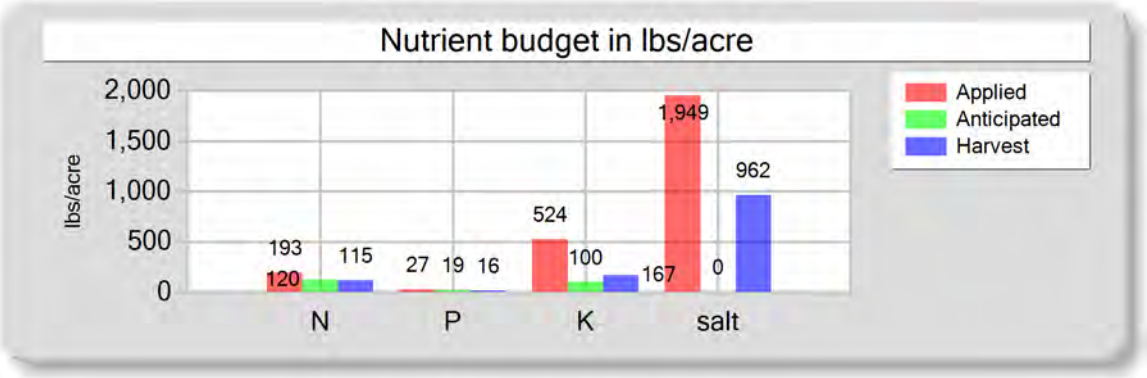
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	1.86	0.00	0.00	138.74	9,576,000.00 *gal*
Application event totals		1.86	0.00	0.00	138.74	

B. NUTRIENT BUDGET

Field 1 - 10/27/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.04	27.31	524.45	1,761.63
Fresh water	2.52	0.00	0.00	187.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.57	27.31	524.45	1,949.39
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	114.50	15.99	167.02	962.18
Nutrient balance	78.06	11.31	357.44	987.21
Applied to removed ratio	1.68	1.71	3.14	2.03

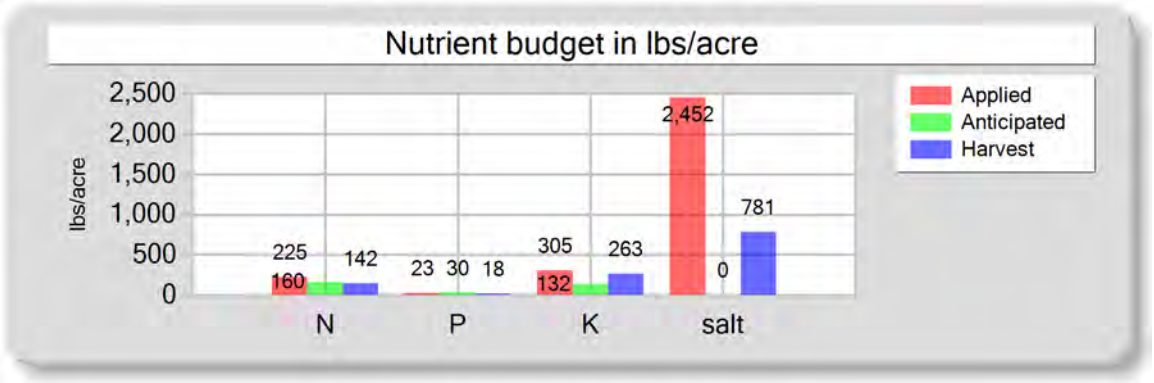
Fresh water applied	
2,340,000.00	*gallons*
86.17	*acre-inches*
6.63	*inches/acre*

Process wastewater applied	
273,500.00	*gallons*
10.07	*acre-inches*
0.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 06/29/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.83	23.11	305.23	1,625.35
Fresh water	11.10	0.00	0.00	826.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.93	23.11	305.23	2,451.50
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	142.30	18.06	263.36	781.15
Nutrient balance	82.63	5.04	41.87	1,670.35
Applied to removed ratio	1.58	1.28	1.16	3.14

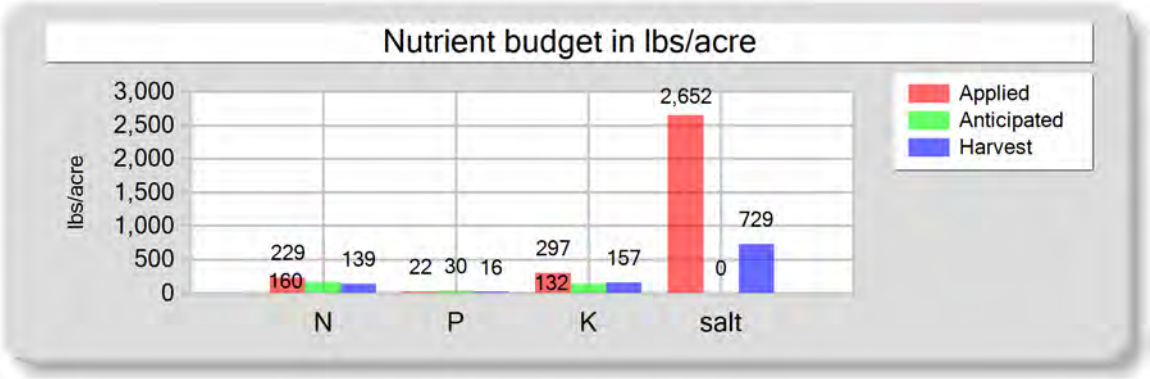
Fresh water applied	
10,296,000.00	*gallons*
379.17	*acre-inches*
29.17	*inches/acre*

Process wastewater applied	
480,000.00	*gallons*
17.68	*acre-inches*
1.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/29/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	200.61	22.16	296.53	1,578.52
Fresh water	14.42	0.00	0.00	1,073.21
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	229.03	22.16	296.53	2,651.73
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	138.77	15.51	157.00	728.81
Nutrient balance	90.26	6.65	139.53	1,922.93
Applied to removed ratio	1.65	1.43	1.89	3.64

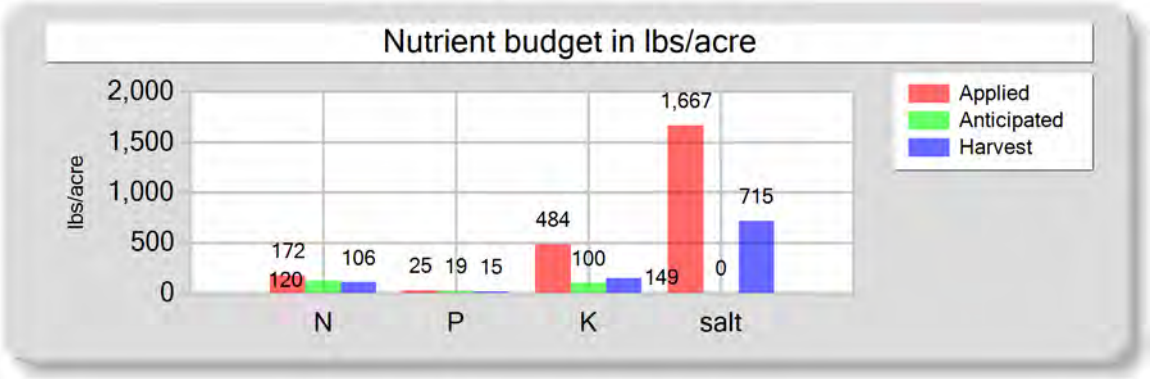
Fresh water applied	
39,096,000.00	*gallons*
1,439.77	*acre-inches*
37.89	*inches/acre*

Process wastewater applied	
1,380,000.00	*gallons*
50.82	*acre-inches*
1.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/27/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	162.79	24.99	484.01	1,500.74
Fresh water	2.23	0.00	0.00	166.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.02	24.99	484.01	1,666.76
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	106.29	15.28	148.67	715.30
Nutrient balance	65.73	9.71	335.34	951.46
Applied to removed ratio	1.62	1.64	3.26	2.33

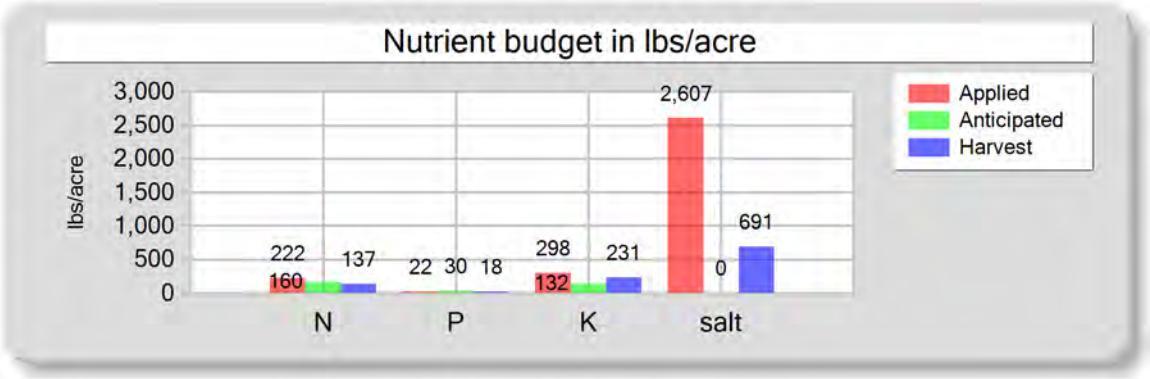
Fresh water applied	
6,048,000.00	*gallons*
222.73	*acre-inches*
5.86	*inches/acre*

Process wastewater applied	
661,000.00	*gallons*
24.34	*acre-inches*
0.64	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/20/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.76	22.34	298.15	1,587.22
Fresh water	13.70	0.00	0.00	1,019.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.46	22.34	298.15	2,606.50
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	136.56	18.00	231.03	691.18
Nutrient balance	85.91	4.34	67.12	1,915.32
Applied to removed ratio	1.63	1.24	1.29	3.77

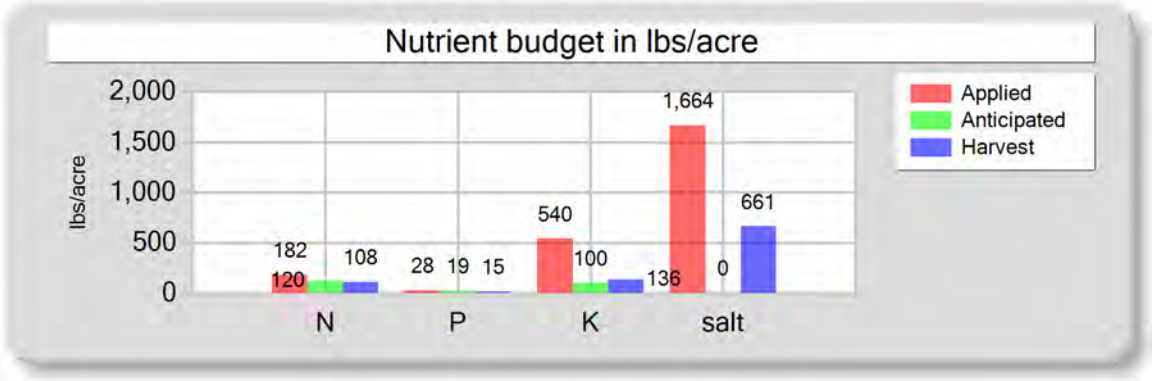
Fresh water applied	
54,720,000.00	*gallons*
2,015.15	*acre-inches*
35.98	*inches/acre*

Process wastewater applied	
2,040,000.00	*gallons*
75.13	*acre-inches*
1.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 10/27/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	172.66	27.56	539.60	1,493.16
Fresh water	2.30	0.00	0.00	170.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	181.96	27.56	539.60	1,663.97
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	107.89	14.75	135.61	661.17
Nutrient balance	74.07	12.81	403.99	1,002.80
Applied to removed ratio	1.69	1.87	3.98	2.52

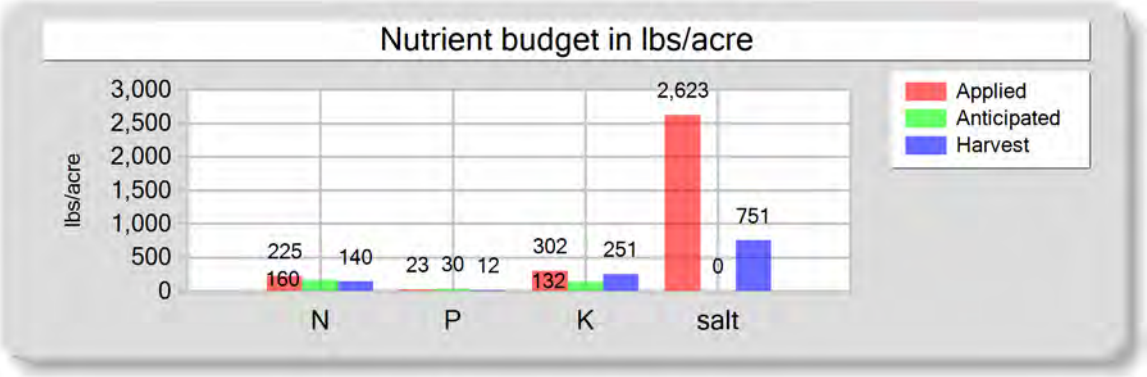
Fresh water applied	
5,076,000.00	*gallons*
186.93	*acre-inches*
6.03	*inches/acre*

Process wastewater applied	
511,000.00	*gallons*
18.82	*acre-inches*
0.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/20/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.20	22.71	301.56	1,605.58
Fresh water	13.67	0.00	0.00	1,017.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.87	22.71	301.56	2,622.62
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	139.76	11.62	251.04	751.02
Nutrient balance	85.11	11.09	50.52	1,871.60
Applied to removed ratio	1.61	1.95	1.20	3.49

Fresh water applied	
70,200,000.00	*gallons*
2,585.23	*acre-inches*
35.91	*inches/acre*

Process wastewater applied	
2,640,000.00	*gallons*
97.22	*acre-inches*
1.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 29.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,900.00	9,200.00	35,300.00	0.00	0.00	0.00	0.00	0.00		33.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	9,210.00	39,900.00	0.00	0.00	0.00	0.00	0.00		40.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

Slurry

Sample and source description: Slurry

Sample date: 08/04/2022 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,500.00	288.00	0.00	0.00	252.00	5,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11,800
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	666.00	344.00	0.00	0.00	76.10	1,330.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9,250.00	8,580
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 01/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	703.00	389.00	0.00	0.00	124.00	1,020.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7,720.00	4,900
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	885.00	588.00	0.00	0.00	118.00	1,270.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10,100.00	6,800
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	543.00	364.00	0.00	0.00	49.20	823.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6,690.00	4,360
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,210.00	490.00	0.00	0.00	49.70	823.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7,160.00	4,120
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

Ag Well

Red House Ag Well

Sample description: Red House Ag Well

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	209.00	125
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

D. SOIL ANALYSES

Field 1

Soil

Sample and source description: Soil

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			24.00				
DL			4.00				

Field 2

Field 2

Soil

Sample and source description: Soil

Sample date: 10/30/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			14.00				
DL			4.00				

Field 3

Soil

Sample and source description: Soil

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			57.00				
DL			4.00				

Field 4

Soil

Sample and source description: Soil

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Field 1 - 10/27/2022: Oats, silage-soft dough

Field 1 - 10/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,090.00	1,130.00	11,800.00		16.50
DL	100.00	100.00	100.00		0.01

Field 1 - 06/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,490.00	570.00	8,310.00		7.80
DL	100.00	100.00	100.00		0.01

Field 2 - 06/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	570.00	5,770.00		6.20
DL	100.00	100.00	100.00		0.01

Field 3 - 10/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,650.00	1,100.00	10,700.00		13.00
DL	100.00	100.00	100.00		0.01

Field 3 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,160.00	680.00	8,730.00		7.75
DL	100.00	100.00	100.00		0.01

Field 4 - 10/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,240.00	990.00	9,100.00		11.80
DL	100.00	100.00	100.00		0.01

Field 4 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,810.00	400.00	8,640.00		7.58
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

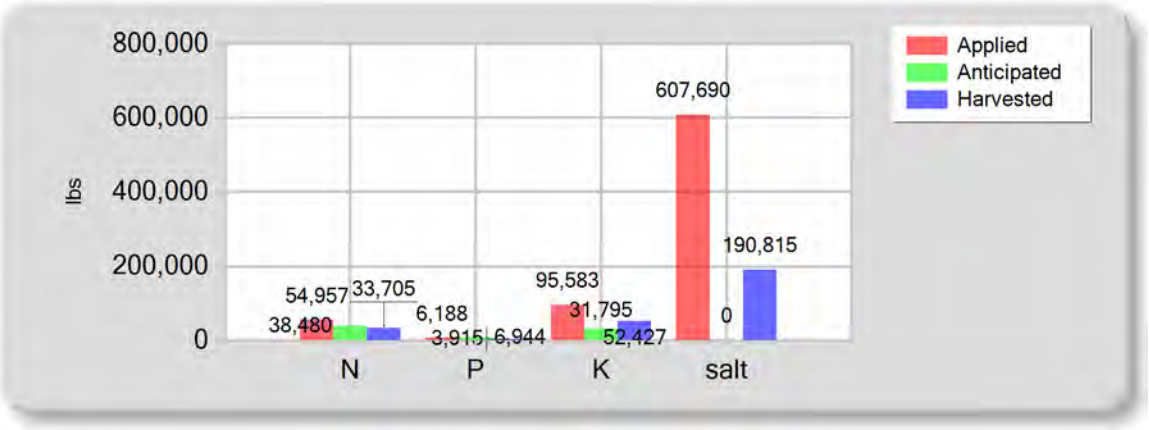
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

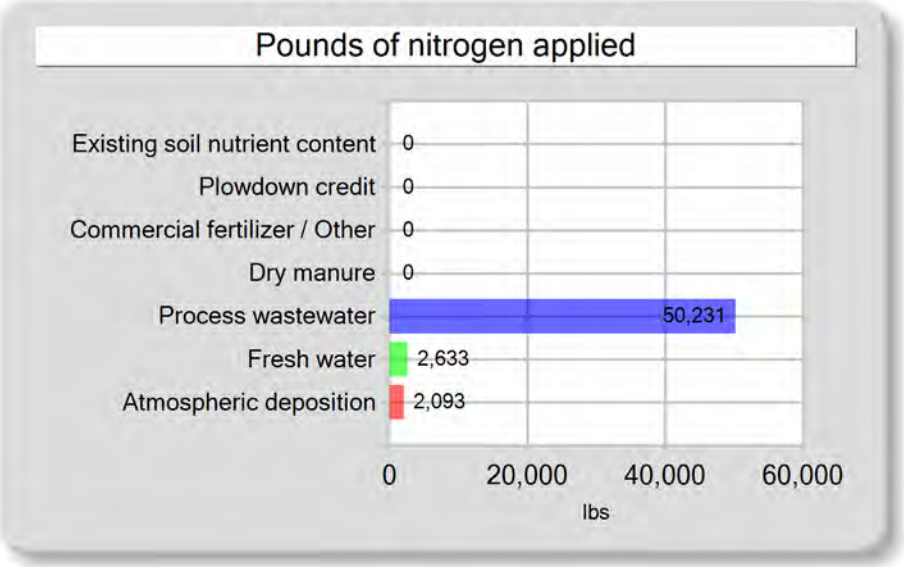
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

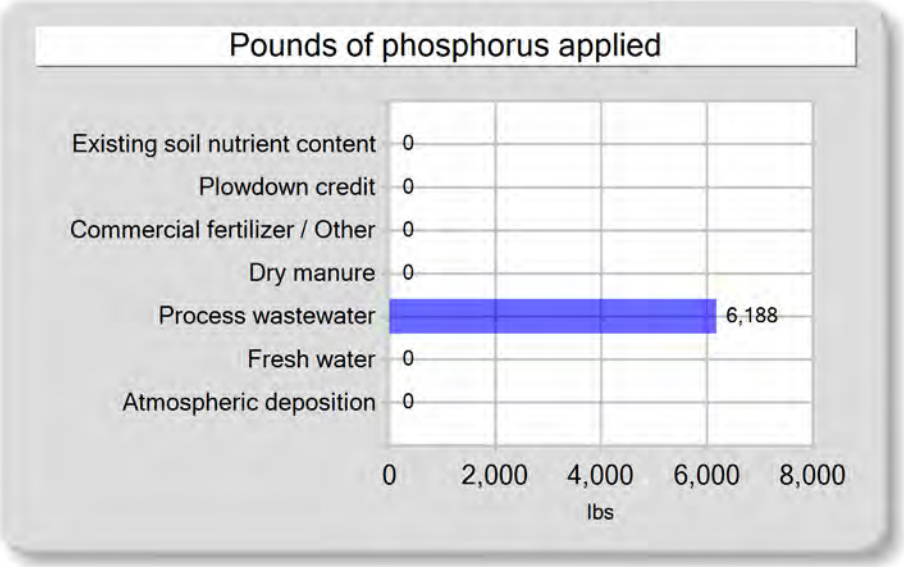
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	50,231.48	6,187.70	95,582.80	411,816.57
Fresh water	2,632.54	0.00	0.00	195,873.84
Atmospheric deposition	2,093.00	0.00	0.00	0.00
Total nutrients applied	54,957.02	6,187.70	95,582.80	607,690.41
Anticipated crop nutrient removal	38,480.00	6,944.40	31,795.20	0.00
Actual crop nutrient removal	33,705.27	3,914.84	52,427.28	190,815.37
Nutrient balance	21,251.75	2,272.86	43,155.52	416,875.04
Applied to removed ratio	1.63	1.58	1.82	3.18

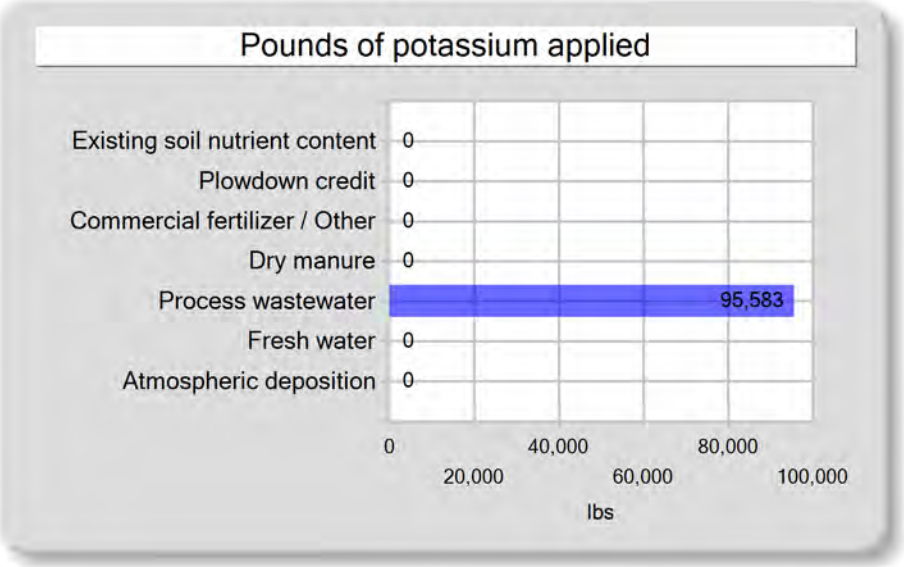
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

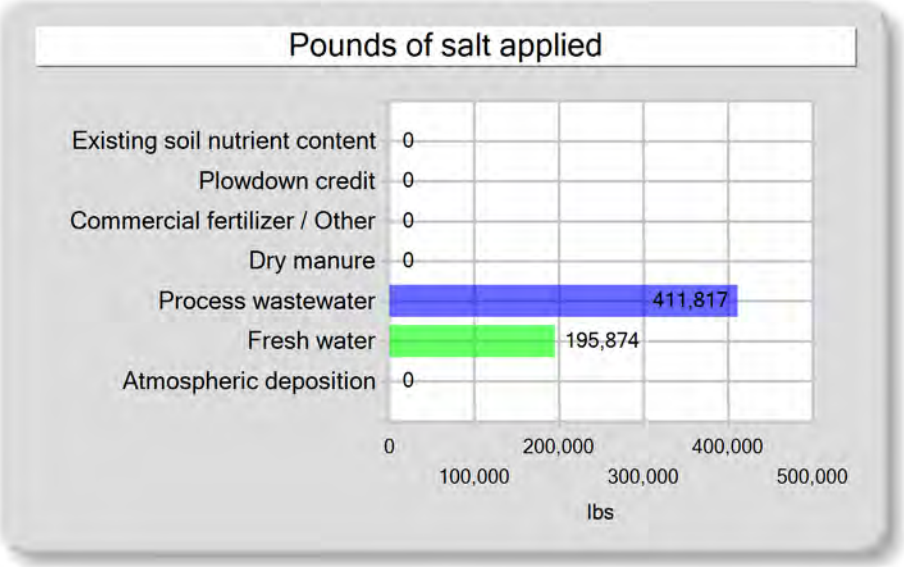


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

During the months of December 2022 to May 2023 no wastewater applications took place as the bottom portions of the cropland were flooded with the heavy rainfall that was received over the winter. During this time wastewater was able to be stored in the recently constructed WWS 2. Only 82 acres of the Oat Silage crop was able to be salvaged to reap a harvest.

Due to the flooding, the Oat Silage crop was severely diminished which resulted in high ratios on Fields 1,3, and 4. The crop yielded an average of 7 tons/acre in contrast to the anticipated 12 tons/acre.

In addition, the Corn Silage crop produced a disappointing yield averaging 14 tons/acre compared to the expected 20 tons per acre. This resulted in high ratios on Fields 1-4, despite nutrient applications being in line with the dairy's Nutrient Management Plan. Farming practices are being evaluated to achieve higher yields from the cropland.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Barbara Garcia	Barbara Garcia
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Barbara Olene Garcia	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
5/10/24	5/10/24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Garcias Registered Jerseys LLC
P.O. Box 544
Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/27/23 11:26 AM
SUBMITTED BY: Adriano
ANALYZED DATE: 12/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy Domestic Well	T3361007A	Electrical Conductivity (EC)	EPA 120.1 and	0.198	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.69	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	126	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.0	100 (T3361001G)	98.2 (T3361001G)	1.81
Electrical Conductivity (EC)	98.8			0.901
Nitrate Nitrogen (NO3-N)	103	104 (T3361008A)	100 (T3361008A)	3.27

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Garcias Registered Jerseys LLC
P.O. Box 544
Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/27/23 11:26 AM
SUBMITTED BY: Adriano
ANALYZED DATE: 12/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Red House Doestic Well	T3361007B	Electrical Conductivity (EC)	EPA 120.1 and	0.691	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	442	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.0	100 (T3361001G)	98.2 (T3361001G)	1.81
Electrical Conductivity (EC)	98.8			0.901
Nitrate Nitrogen (NO3-N)	103	104 (T3361008A)	100 (T3361008A)	3.27

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ELAP Certificate No.: 2714

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40 N. East Street, Woodland, CA 95776
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Compliance Analysis Report

Customer
Garcias Registered Jerseys LLC
P.O. Box 544
Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/27/23 11:26 AM
SUBMITTED BY: Adriano
ANALYZED DATE: 12/27/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 18.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Red House Ag Well	T3361007C	Electrical Conductivity (EC)	EPA 120.1 and	0.209	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.68	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	125	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	1.68	mg/L
		Soluble Salts (SALT-SOL)	Calculation	133	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	99.0	100 (T3361001G)	98.2 (T3361001G)			1.81
Electrical Conductivity (EC)	98.8			0.669 (T3361001I)	0.663 (Dup)	0.901
Nitrate Nitrogen (NO3-N)	103	104 (T3361008A)	100 (T3361008A)			3.27
Total Dissolved Solids (TDS)	104			560 (T3362017B)	595 (Dup)	6.06
Total Nitrogen (TN)	102			0.280 (T3353003G)	0.280 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Garcia's Registered Jerseys
Mailing address: P.O. Box 544 Denair, Ca 95316
(not facility address)
Phone#: (209) 321-5603 / (209) 202-4853

Authorized Copy Released To: F&R Ag

Sampled By: Adriano Garcia
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12/27/23	11:00	Dairy Domestic Well		11.6°C		X				
2	12/27/23	11:00	Red House Domestic Well		11.4°		X				
3	12/27/23	11:00	Red House Ag Well		18.6°C						H7
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	Adriano G					
2nd		DENELE ANALYTICAL, INC.	12-27-23 11:26			T 3361007

Required to be submitted to any agency? (YES / NO)